

# UNIT-I – ASP.NET Controls

ASP.NET uses five types of web controls, which are:

- HTML controls
- HTML Server controls
- ASP.NET Server controls
- ASP.NET Ajax Server controls

## User controls and custom controls

ASP.NET server controls are the primary controls used in ASP.NET. These controls can be grouped into the following categories:

**Validation controls** - These are used to validate user input and they work by running client-side script.

**Data source controls** - These controls provides data binding to different data sources.

**Data view controls** - These are various lists and tables, which can bind to data from data sources for displaying.

**Personalization controls** - These are used for personalization of a page according to the user preferences, based on user information.

**Login and security controls** - These controls provide user authentication.

**Master pages** - These controls provide consistent layout and interface throughout the application.

**Navigation controls** - These controls help in navigation. For example, menus, tree view etc.

**Rich controls** - These controls implement special features. For example, AdRotator, FileUpload, and Calendar control.

## ASP.NET Web Forms Server Controls

ASP.NET provides web forms controls that are used to create HTML components. These controls are categories as server and client based. The following table contains the server controls for the web forms.

| Control Name | Applicable Events                                                                                                                                      | Description                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Label        | None                                                                                                                                                   | It is used to display text on the HTML page.                                                                                           |
| TextBox      | TextChanged                                                                                                                                            | It is used to create a text input in the form.                                                                                         |
| Button       | Click, Command                                                                                                                                         | It is used to create a button.                                                                                                         |
| LinkButton   | Click, Command                                                                                                                                         | It is used to create a button that looks similar to the hyperlink.                                                                     |
| ImageButton  | Click                                                                                                                                                  | It is used to create an imagesButton. Here, an image works as a Button.                                                                |
| Hyperlink    | None                                                                                                                                                   | It is used to create a hyperlink control that responds to a click event.                                                               |
| DropDownList | SelectedIndexChanged                                                                                                                                   | It is used to create a dropdown list control.                                                                                          |
| ListBox      | SelectedIndexCnhaged                                                                                                                                   | It is used to create a ListBox control like the HTML control.                                                                          |
| DataGrid     | CancelCommand, EditCommand, DeleteCommand, ItemCommand, SelectedIndexChanged, PageIndexChanged, SortCommand, UpdateCommand, ItemCreated, ItemDataBound | It used to create a frid that is used to show data. We can also perform paging, sorting, and formatting very easily with this control. |
| DataList     | CancelCommand, EditCommand, DeleteCommand, ItemCommand, SelectedIndexChanged, UpdateCommand, ItemCreated, ItemDataBound                                | It is used to create datalist that is non-tabular and used to show data.                                                               |

|                 |                                                     |                                                                                              |
|-----------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------|
| CheckBox        | CheckChanged                                        | It is used to create checkbox.                                                               |
| CheckBoxList    | SelectedIndexChanged                                | It is used to create a group of check boxes that all work together.                          |
| RadioButton     | CheckChanged                                        | It is used to create radio button.                                                           |
| RadioButtonList | SelectedIndexChanged                                | It is used to create a group of radio button controls that all work together.                |
| Image           | None                                                | It is used to show image within the page.                                                    |
| Panel           | None                                                | It is used to create a panel that works as a container.                                      |
| Placeholder     | None                                                | It is used to set placeholder for the control.                                               |
| Calendar        | SelectionChanged,<br>VisibleMonthChanged, DayRender | It is used to create a calendar. We can set the default date, move forward and backward etc. |
| AdRotator       | AdCreated                                           | It allows us to specify a list of ads to display. Each time the user re-displays the page.   |
| Table           | None                                                | It is used to create table.                                                                  |
| XML             | None                                                | It is used to display XML documents within the HTML.                                         |

|         |      |                                                                                                                                     |
|---------|------|-------------------------------------------------------------------------------------------------------------------------------------|
| Literal | None | It is like a label in that it displays a literal, but allows us to create new literals at runtime and place them into this control. |
|---------|------|-------------------------------------------------------------------------------------------------------------------------------------|

## ASP.NET Web Forms Label

This control is used to display textual information on the web forms. It is mainly used to create caption for the other controls like: textbox.

This is server side control, asp provides own tag to create label. The example is given below.

```
< asp:LabelID="Label1" runat="server" Text="Label" ></asp:Label>
```

This control has its own properties that are tabled below.

| Property    | Description                                        |
|-------------|----------------------------------------------------|
| AccessKey   | It is used to set keyboard shortcut for the label. |
| TabIndex    | The tab order of the control.                      |
| BackColor   | It is used to set background color of the label.   |
| BorderColor | It is used to set border color of the label.       |
| BorderWidth | It is used to set width of border of the label.    |
| Font        | It is used to set font for the label text.         |
| ForeColor   | It is used to set color of the label text.         |
| Text        | It is used to set text to be shown for the label.  |
| ToolTip     | It displays the text when mouse is over the label. |

|         |                                           |
|---------|-------------------------------------------|
| Visible | To set visibility of control on the form. |
| Height  | It is used to set height of the control.  |
| Width   | It is used to set width of the control.   |

### ASP.NET Web Forms TextBox

This is an input control which is used to take user input. To create TextBox either we can write code or use the drag and drop facility of visual studio IDE.

This is server side control, asp provides own tag to create it. The example is given below.

```
< asp:TextBoxID="TextBox1" runat="server" ></asp:TextBox>
```

Server renders it as the HTML control and produces the following code to the browser.

```
<input name="TextBox1" id="TextBox1" type="text">
```

This control has its own properties that are tabled below.

| Property    | Description                                          |
|-------------|------------------------------------------------------|
| AccessKey   | It is used to set keyboard shortcut for the control. |
| TabIndex    | The tab order of the control.                        |
| BackColor   | It is used to set background color of the control.   |
| BorderColor | It is used to set border color of the control.       |
| BorderWidth | It is used to set width of border of the control.    |
| Font        | It is used to set font for the control text.         |
| ForeColor   | It is used to set color of the control text.         |

|           |                                                                     |
|-----------|---------------------------------------------------------------------|
| Text      | It is used to set text to be shown for the control.                 |
| ToolTip   | It displays the text when mouse is over the control.                |
| Visible   | To set visibility of control on the form.                           |
| Height    | It is used to set height of the control.                            |
| Width     | It is used to set width of the control.                             |
| MaxLength | It is used to set maximum number of characters that can be entered. |
| Readonly  | It is used to make control readonly.                                |

### ASP.NET Web Forms Button

This control is used to perform events. It is also used to submit client request to the server. To create Button either we can write code or use the drag and drop facility of visual studio IDE.

This is a server side control and asp provides own tag to create it. The example is given below.

```
<asp:ButtonID="Button1" runat="server" Text="Submit" BorderStyle="Solid" ToolTip="Submit"/>
```

Server renders it as the HTML control and produces the following code to the browser.

```
<input name="Button1" value="Submit" id="Button1" title="Submit" style="border-style:Solid;" type="submit">
```

This control has its own properties that are tabled below.

| Property  | Description                                          |
|-----------|------------------------------------------------------|
| AccessKey | It is used to set keyboard shortcut for the control. |
| TabIndex  | The tab order of the control.                        |

|             |                                                      |
|-------------|------------------------------------------------------|
| BackColor   | It is used to set background color of the control.   |
| BorderColor | It is used to set border color of the control.       |
| BorderWidth | It is used to set width of border of the control.    |
| Font        | It is used to set font for the control text.         |
| ForeColor   | It is used to set color of the control text.         |
| Text        | It is used to set text to be shown for the control.  |
| ToolTip     | It displays the text when mouse is over the control. |
| Visible     | To set visibility of control on the form.            |
| Height      | It is used to set height of the control.             |
| Width       | It is used to set width of the control.              |

### **ASP.NET Web Forms HyperLink**

It is a control that is used to create a hyperlink. It responds to a click event. We can use it to refer any web page on the server.

To create HyperLink either we can write code or use the drag and drop facility of visual studio IDE. This control is listed in the toolbox.

This is a server side control and ASP.NET provides own tag to create it. The example is given below.

```
<asp:HyperLinkID="HyperLink1" runat="server" Text="JavaTpoint" NavigateUrl="www.javatpoint.com" ></asp:HyperLink>
```

Server renders it as the HTML control and produces the following code to the browser.

```
<a id="HyperLink1" href="wwwmjcollege.com">JavaTpoint</a>
```

This control has its own properties that are tabled below.

| Property    | Description                                          |
|-------------|------------------------------------------------------|
| AccessKey   | It is used to set keyboard shortcut for the control. |
| TabIndex    | The tab order of the control.                        |
| BackColor   | It is used to set background color of the control.   |
| BorderColor | It is used to set border color of the control.       |
| BorderWidth | It is used to set width of border of the control.    |
| Font        | It is used to set font for the control text.         |
| ForeColor   | It is used to set color of the control text.         |
| Text        | It is used to set text to be shown for the control.  |
| ToolTip     | It displays the text when mouse is over the control. |
| Visible     | To set visibility of control on the form.            |
| Height      | It is used to set height of the control.             |
| Width       | It is used to set width of the control.              |
| NavigateUrl | It is used to set navigate URL.                      |
| Target      | Target frame for the navigate url.                   |

## ASP.NET LinkButton



It is a server web control that acts as a hyperlink. It is used to display a hyperlink-style button control on the web page. ASP.NET provides a tag to create LinkButton and has following syntax.

#### ASP.NET LinkButton Syntax

```
<asp:LinkButton
    AccessKey="string"
    ID="string"
    OnClick="Click event handler"
    OnClientClick="string"
    OnCommand="Command event handler"
    OnDataBinding="DataBinding event handler"
    OnDisposed="Disposed event handler"
    OnInit="Init event handler"
    OnLoad="Load event handler"
    OnPreRender="PreRender event handler"
    OnUnload="Unload event handler"
   PostBackUrl="uri"
    runat="server"
/>
```

#### ASP.NET DropDownList

The DropDownList is a web server control which is used to create an HTML Select component. It allows us to select an option from the dropdown list. It can contain any number of items

ASP.NET provides a tag to create DropDownList for web application. The following is the Syntax of DropDownList tag.

```
<asp:DropDownList id="DropDownList1" runat="server"
    DataSource="<% databindingexpression %>"
    DataTextField="DataSourceField"
```

DataValueField="DataSourceField"

AutoPostBack="True|False"

OnSelectedIndexChanged="OnSelectedIndexChangedMethod">

<asp:ListItem value="value" selected="True|False">

Text

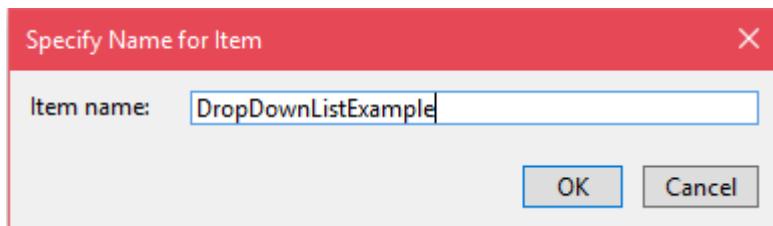
</asp:ListItem>

</asp:DropDownList>

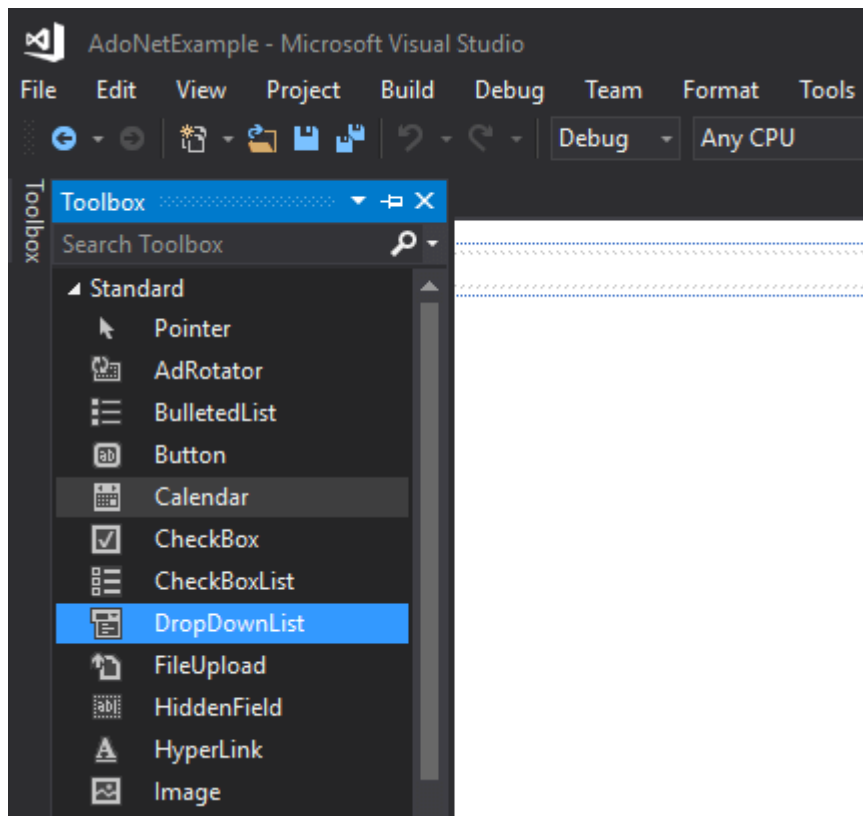
## ASP.NET DropDownList Example

### Create a Web Form

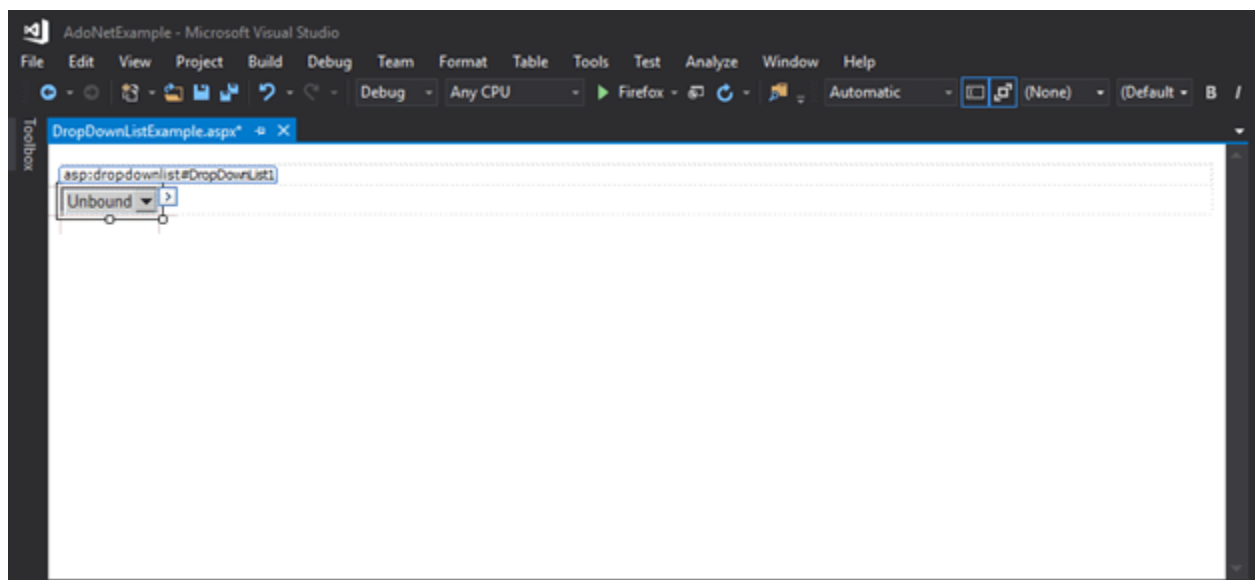
Add a new form by specifying its name.

A screenshot of a Windows-style dialog box titled "Specify Name for Item" with a red header bar and a close button (X) in the top right corner. The dialog contains a label "Item name:" followed by a text input field. The input field contains the text "DropDownListExample". At the bottom right of the dialog are two buttons: "OK" and "Cancel".

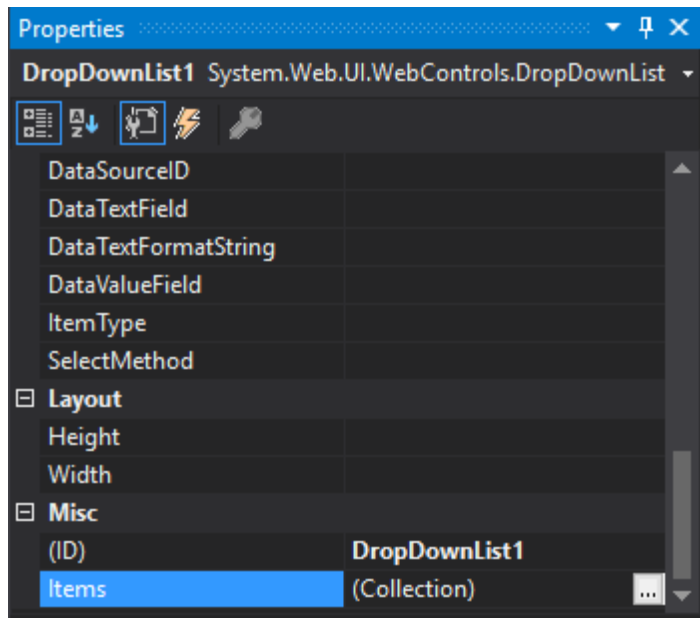
Initially, it is an empty form. Now, we will add a new DropDownList by dragging it from the toolbox.



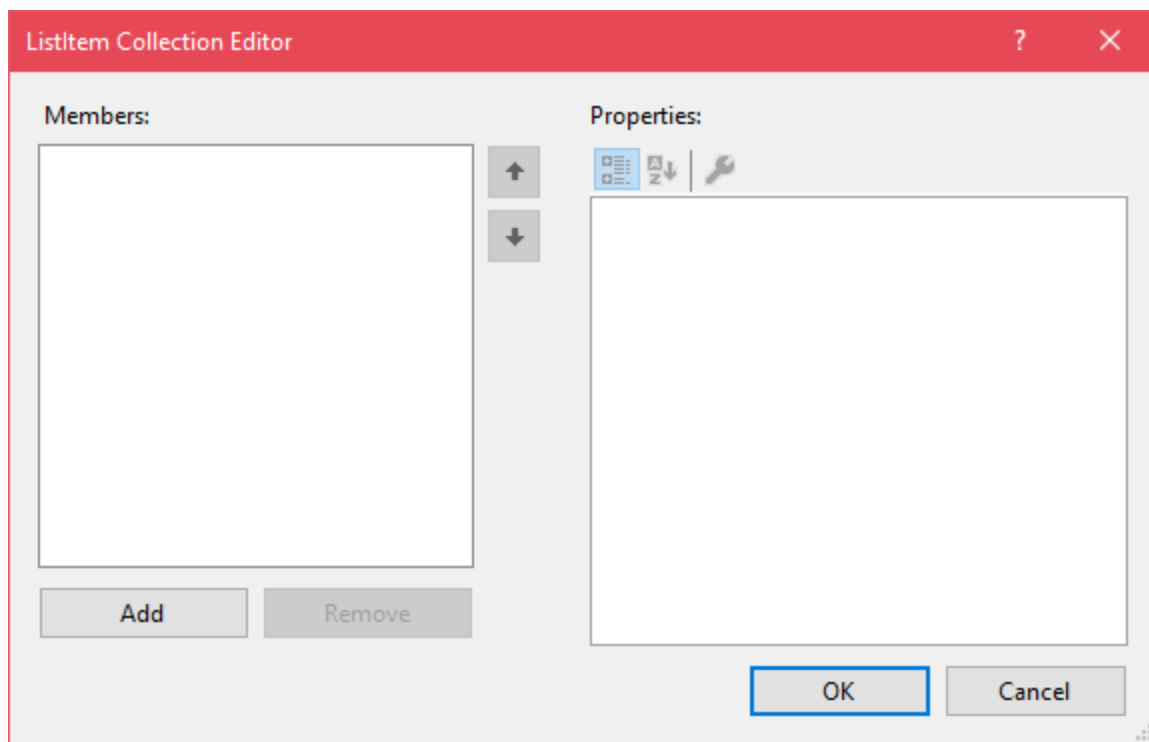
After dragging, our web form looks like the below.



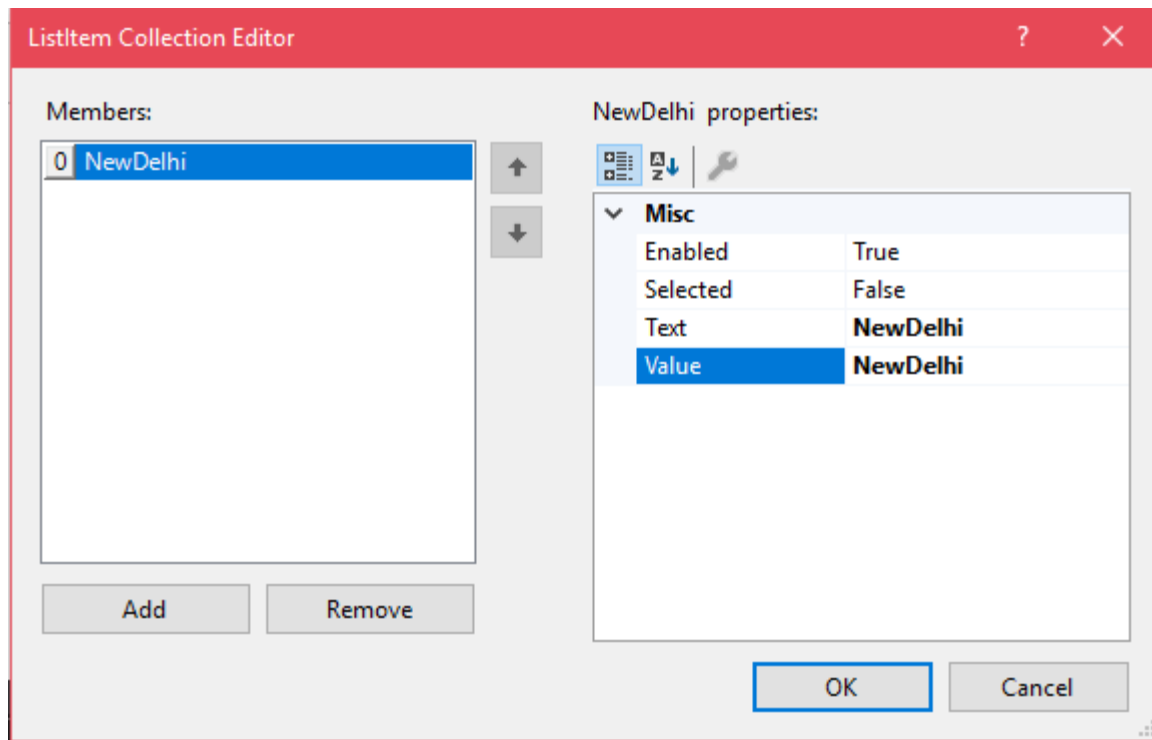
Now, to add items to the list, visual studio provides Items property where we can add items. The property window looks like this.



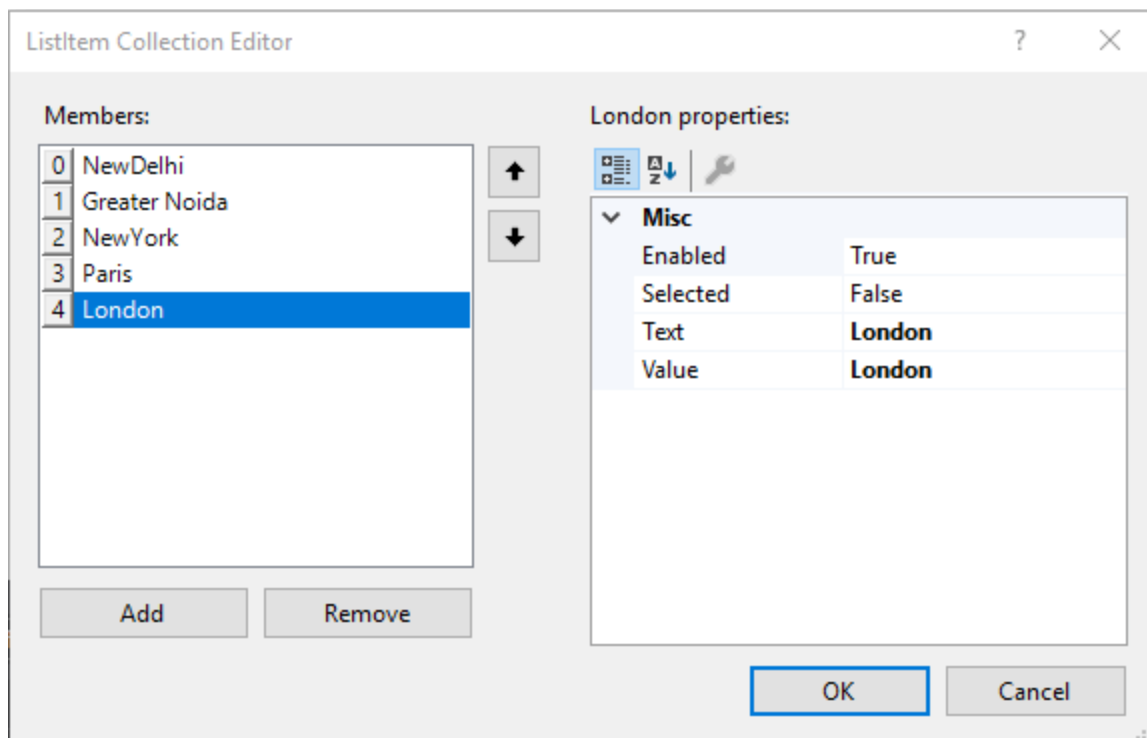
Click on the items (collection) and it will pop up a new window as given below. Initially, it does not have any item. It provides add button to add new items to the list.



Adding item to the DropDownList, by providing values to the Text and Value properties.



We have added more items to it and now, it looks like the following.



After clicking OK.

## ASP.NET Web Forms CheckBox

- It is used to get multiple inputs from the user. It allows user to select choices from the set of choices.
- It takes user input in yes or no format. It is useful when we want multiple choices from the user.
- To create CheckBox we can drag it from the toolbox in visual studio.

This is a server side control and ASP.NET provides own tag to create it. The example is given below.

```
< asp:CheckBox ID="CheckBox2" runat="server" Text="J2EE"/>
```

Server renders it as the HTML control and produces the following code to the browser.

```
< input id="CheckBox2" type="checkbox" name="CheckBox2" /><label for="CheckBox2">J2EE</label>
```

This control has its own properties that are tabled below.

| Property    | Description                                          |
|-------------|------------------------------------------------------|
| AccessKey   | It is used to set keyboard shortcut for the control. |
| TabIndex    | The tab order of the control.                        |
| BackColor   | It is used to set background color of the control.   |
| BorderColor | It is used to set border color of the control.       |
| BorderWidth | It is used to set width of border of the control.    |
| Font        | It is used to set font for the control text.         |
| ForeColor   | It is used to set color of the control text.         |
| Text        | It is used to set text to be shown for the control.  |
| ToolTip     | It displays the text when mouse is over the control. |

|         |                                                                    |
|---------|--------------------------------------------------------------------|
| Visible | To set visibility of control on the form.                          |
| Height  | It is used to set height of the control.                           |
| Width   | It is used to set width of the control.                            |
| Checked | It is used to set check state of the control either true or false. |

### ASP.NET Web Forms RadioButton

It is an input control which is used to takes input from the user. It allows user to select a choice from the group of choices.

To create RadioButton we can drag it from the toolbox of visual studio.

This is a server side control and ASP.NET provides own tag to create it. The example is given below.

```
<asp:RadioButtonID="RadioButton1" runat="server" Text="Male" GroupName="gender"/>
```

Server renders it as the HTML control and produces the following code to the browser.

```
<input id="RadioButton1" type="radio" name="gender" value="RadioButton1" /><label for="RadioButton1">Male</label>
```

This control has its own properties that are tabled below.

| Property    | Description                                          |
|-------------|------------------------------------------------------|
| AccessKey   | It is used to set keyboard shortcut for the control. |
| TabIndex    | The tab order of the control.                        |
| BackColor   | It is used to set background color of the control.   |
| BorderColor | It is used to set border color of the control.       |
| BorderWidth | It is used to set width of border of the control.    |

|           |                                                      |
|-----------|------------------------------------------------------|
| Font      | It is used to set font for the control text.         |
| ForeColor | It is used to set color of the control text.         |
| Text      | It is used to set text to be shown for the control.  |
| ToolTip   | It displays the text when mouse is over the control. |
| Visible   | To set visibility of control on the form.            |
| Height    | It is used to set height of the control.             |
| Width     | It is used to set width of the control.              |
| GroupName | It is used to set name of the radio button group.    |

### **The Image Control.**

The image control is used for displaying images on the web page, or some alternative text, if the image is not available.

Basic syntax for an image control:

```
<asp:Image ID="Image1" runat="server">
```

It has the following important properties:

| Properties    | Description                                             |
|---------------|---------------------------------------------------------|
| AlternateText | Alternate text to be displayed in absence of the image. |
| ImageAlign    | Alignment options for the control.                      |
| ImageUrl      | Path of the image to be displayed by the control.       |